GEAD: Explainable Deep Learning-Based Anomaly Detection for Network Security

Implementation • Explainability • Intrusion Detection

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Outline

1. Motivation & Problem Statement
2. GEAD Architecture
3. Implementation & Experimental Setup
4. Experimental Results
5. Conclusion

The Problem: Why Traditional IDS Systems Are Hard to Trust

Modern Intrusion Detection Systems (IDS) increasingly use deep learning models for anomaly detection.

However, deep neural networks suffer from:

1. **Black-box Decision Making**
   Models detect anomalies but cannot explain why traffic is malicious.

   Security analysts cannot easily verify predictions.

2. **Lack of Interpretability**
   Traditional deep autoencoders provide only anomaly scores.

   No human-readable security rules are generated.

3. **Security & Trust Issues**
   Enterprise IDS systems require explainability for:

GEAD Framework Architecture

GEAD combines neural anomaly detection with explainable decision trees.

1. **Network Traffic Collection**
   CICIDS2017 dataset used for intrusion detection.
2. **Autoencoder-based Anomaly Detection**
   Autoencoder trained only on benign traffic.

   Reconstruction error used as anomaly score.
3. **Root Regression Tree Generation**
   Regression tree approximates neural network behavior.
4. **Low-confidence Leaf Detection**
   Identifies uncertain prediction regions.
5. **Fine-grained Rule Generation**

Implementation & Experimental Setup

Development Environment

- Google Colab
- Python 3.12
- PyTorch
- Scikit-learn
- NumPy, Pandas, Matplotlib

Dataset

CICIDS2017 Network Intrusion Dataset

- Benign traffic
- DoS attacks
- Brute force attacks
- Web attacks

Experimental Results

1. Autoencoder Performance

Successfully learned benign network traffic behavior using reconstruction error.

2. Root Regression Tree Generation

- 41 leaf nodes generated
- Maximum depth $= 12$
- 24 low-confidence leaves identified

3. Data Augmentation

Generated:

$$22611 \to 201430$$

augmented explainability samples.

Fine-grained Rule Generation & Visualizations

4. Fine-grained Rule Generation

Successfully produced:

$$47$$

interpretable local security rules.

Visualizations Generated

- ROC Curve
- Reconstruction Error Distribution
- Regression Tree Statistics
- Low-confidence Leaf Analysis

Result: GEAD successfully generated interpretable anomaly explanations.

Conclusion

GEAD provides an explainable intrusion detection framework by combining deep anomaly detection with interpretable rule extraction.

Key Achievements

1. Successfully reproduced the anomaly detection pipeline.
2. Generated interpretable fine-grained security rules.
3. Demonstrated explainability for deep learning-based IDS systems.
4. Identified compatibility challenges with modern sklearn tree internals.

Future Scope

- Real-time IDS deployment
- Modern sklearn compatibility support
- Integration with enterprise SIEM systems